2.2 Million Stories

How the New York Times connected countries across 25 years

Process Book

COM-480 Data Visualization — EPFL

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Spring 2026

Process Book - 2.2 Million Stories: Mapping the NYT

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Week 1 - February 16-22, 2026

Starting point and motivation

I wanted to explore how the New York Times covers the world by analyzing geographic patterns across 2.2M articles (2000-2023). The NYT frequently mentions countries in its headlines, which makes it possible to extract geographic attention patterns from the dataset without full article text. The dataset contains headlines, abstracts, snippets, keywords, bylines, and section labels.

Data pipeline

I converted the raw CSV (4.3 GB) to SQLite with a date index, which made all subsequent work fast and modular. Every new analysis reads from the same database and writes pre-computed JSON files that the browser loads directly, with no server-side processing.

World map - country mentions over time

The first visualization is an animated choropleth showing per-year country mention frequency. I extract countries using a dictionary of ~200 names and aliases compiled into a single regex. About 31% of articles mention at least one country, with 150 countries detected in total. A log color scale turned out to be necessary because US mentions (~12k/year) are orders of magnitude above smaller countries.

Design and technical choices

I chose vanilla JS and D3.js v7 because I like working without frameworks - it keeps things simple and gives full control. The site uses static hosting only. The main challenges this week were handling the CSV scale, and choosing the right color scale for the map.

AI disclosure

- CSV-to-SQLite conversion was generated with AI assistance
- The regex-based country extraction algorithm was developed with AI support

Week 2 - February 23 - March 1, 2026

Bar chart race - a short detour

I briefly explored an animated bar chart showing the 15 most frequent words per year, but quickly realized it did not fit the main theme of geographic country analysis. I decided to focus entirely on map-based visualizations going forward.

US city bubble map

I added a bubble map of US city mentions using an AlbersUSA projection. The data source is the glocations field in the NYT keyword metadata - structured tags like “Los Angeles (Calif)” or “Manhattan (NYC)”. The preprocessing maps ~200 of these variants to canonical cities with lat/lon coordinates, rolling up NYC boroughs and neighborhoods into a single entry.

Out of 2.2M articles, 188k have US city tags across 160 unique cities. New York City dominates at 120k mentions, followed by Washington DC (15k), Los Angeles (5.7k), and Chicago (4.3k). The map makes the NYT’s geographic concentration immediately visible - heavy coverage in the Northeast corridor and

California, with sparse coverage elsewhere. In dense regions like NYC, LA, and SF, the bubbles overlap heavily and are hard to distinguish. I planned to later add metro sub-zooms into these areas to make them more readable.

The original sketch from milestone 2 translated surprisingly well to the actual implementation - I could more or less bring the sketch directly to the website.

AI disclosure

- The US city dictionary mapping NYT glocations variants to canonical city names was built with AI assistance
- Database query optimization for extracting glocations counts was AI-assisted

Week 3 - March 2-8, 2026

Grid cartogram - average article word count by country

I wanted to create a different kind of map - the NYT itself often uses tile grid cartograms in their own visualizations, so it felt fitting. The tile grid gives every country an equal-sized square arranged in a layout that loosely preserves geographic adjacency, which avoids the problem of small countries being invisible on a geographic map. I used tile positions from the Tile-Grid-Map repository, which provides grid coordinates for 192 countries.

Each cell shows the average article word count for that country, colored on a sequential scale with a gradient color legend at the bottom. Hover tooltips show the country name, average word count, and total article count. This feature was later dismissed as it did not fit nicely into the scroll-driven flow of the final visualization.

Small multiples - 25 yearly maps

I also experimented with a small multiples approach: 25 tiny choropleth maps, one per year, arranged in a grid. The idea was to let the user spot geographic coverage shifts over time at a glance. While it worked as a proof of concept, it consumed too much vertical space and the individual maps were too small to compare meaningfully. This led to the trend arrow approach in week 4. In retrospect, dismissing small multiples was the right call - the final storytelling approach relies on the map always being the same familiar canvas, with different data layers applied on top.

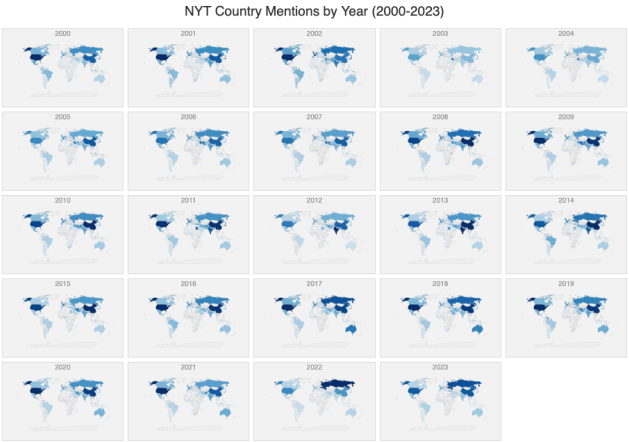
AI disclosure

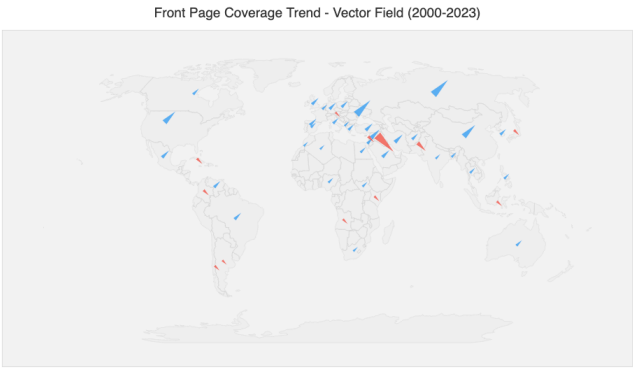
- The grid cartogram layout and D3.js cell rendering were developed with AI assistance
- The color legend implementation was AI-assisted

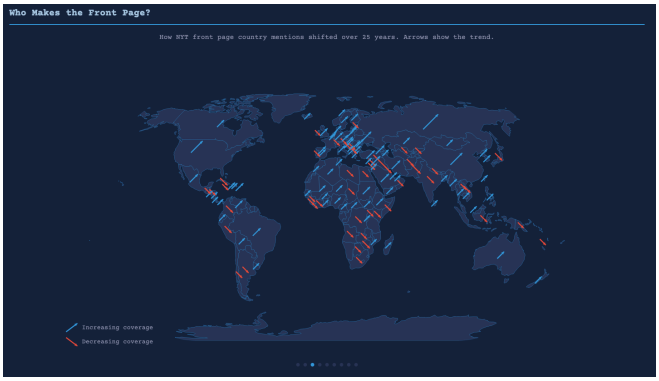
Week 4 - March 9-15, 2026

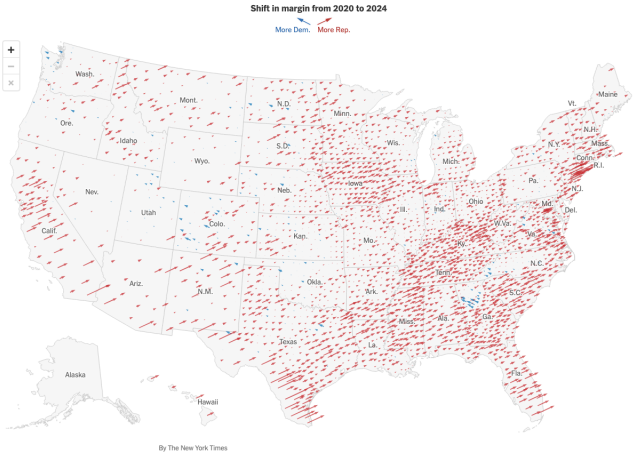
Coverage trend map - replacing small multiples with direction arrows

The 25-tile small multiples grid from week 3 had problems: it consumed too much vertical space, and comparing subtle year-over-year shifts across tiny maps was difficult. I replaced it with a single map where each country displays a directional arrow showing whether its Front Page coverage increased or decreased over 2000-2023.









Top left: Small multiples POC. Top right: Vector field POC. Bottom left: Final trend arrow map – the dark mode really adds a professional glow to it. Bottom right: NYT election arrow map that inspired the approach.

The approach is inspired by New York Times election graphics, which use arrow maps to show shifts in voting patterns – arrows on a map are an effective way to encode direction and magnitude of change without requiring the reader to compare multiple snapshots.

For each country, I compute a linear regression slope across 25 data points. The slope sign determines the arrow direction: positive slopes point up-right (increasing), negative slopes point down-right (decreasing). Arrow length is scaled from the absolute slope.

The result shows roughly what you would expect: Iraq and Afghanistan have declining arrows (peak coverage during the wars, then fading), while countries like Ukraine and China show increasing coverage. The single-map format is far more scannable and communicates the story more directly.

Co-occurrence map – first version

I also started working on the co-occurrence map, which shows which countries are mentioned together in the same NYT articles. The initial version lets the user click a country to see arc connections to the countries most frequently co-mentioned with it. This became the main focus of the following weeks.

AI disclosure

- The trend computation (linear regression) was AI-assisted
- Arrow positioning and SVG marker placement were buggy initially and fixed with AI assistance

Week 5 - March 16-22, 2026

Dark mode

I switched the page to a permanent dark theme. The dark palette immediately improved visual coherence - the map visualizations now blend into the page rather than sitting in conspicuous boxes.

Co-occurrence map - automated exploration animation

I turned the co-occurrence map into a fully automated animation loop: a random country is selected and highlighted, connection arcs draw out one by one toward co-mentioned countries, and then a travel line moves to the next country. The cycle repeats indefinitely. Getting the timing right was the most time-consuming part - early versions switched too abruptly, and the staged approach where arcs stay visible during the travel line was the key insight. This auto-mode was later dismissed as it decreases interactivity and does not let the user explore at their own pace.

Background dot map - the landing page

I added a fixed-position SVG overlay that serves as the landing page. Country mention dots continuously spawn and fade across the map - each dot appears at a random point inside its country, draws a thin line upward, and typewriters the country name. The idea is that it resembles a sneak peek of what is coming while being a nice visual touch before the user enters the actual visualizations.

Scroll-snap and layout

I added scroll-snap globally so each section fills the viewport. This was later replaced by a more tile-like approach where the page stays more fixed, resulting in a more app-like feeling where the user cycles through views rather than scrolling between sections.

AI disclosure

- Dark mode migration was AI-assisted
- Co-occurrence autoplay algorithm was AI-assisted
- Background dot map (spawn/fade lifecycle, geographic sampling, overlay positioning) was AI-assisted

Week 6 - March 23-29, 2026

Typewriter effect

I added a typewriter animation to all text headings. The header title, subtitles, and section titles all type out character by character. The effect re-triggers on each scroll - when a section leaves the viewport, its title is cleared, and when it re-enters, the typewriter runs again.

To make the effect feel more human, I added a typo simulation. I once read an article on Hacker News where this idea was introduced, but I could not find it again. Each character has a small probability of being mistyped using a nearby-key map. When a typo occurs, the wrong character appears briefly, gets deleted, and the correct character is typed - mimicking a real person catching and correcting a mistake.

I applied the same approach to the background dot map’s country name labels, where each label types out letter by letter after the connecting line finishes drawing.

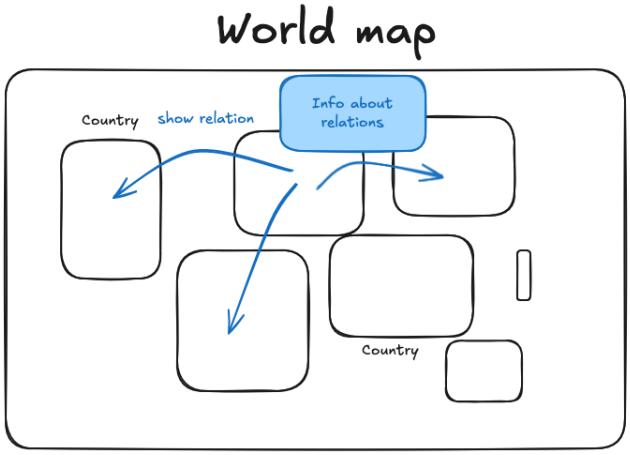
Co-occurrence map - hover interaction

I reworked the co-occurrence map from the automated animation to a hover-triggered interaction. The map now responds to the user hovering over a country, drawing connection arcs to the top 10 co-mentioned countries. When the mouse leaves, everything fades back. I added a generation counter to prevent animation interleaving when the user moves quickly between countries.

The tooltip was expanded into a side panel showing the country name, total headline mentions, an animated sparkline of coverage over time, and a ranked list of the top co-mentioned countries.

Front page map and bubble map improvements

The trend arrow map and US city bubble map both received animated sparkline tooltips showing yearly coverage percentages. The bubble map now dims all other bubbles when hovering over a city, creating a spotlight effect.



Left: Original sketch from milestone 2. Right: Final implementation with hover-triggered arcs and connection panel.

Data normalization

Investigating the trend arrows revealed that almost every country showed decreasing coverage. This turned out to be because the NYT published roughly 110,000 articles per year in the early 2000s but only around 50,000 in recent years. I updated all preprocessing to normalize counts as a percentage of total articles per year. For consistency, dates 2024 and 2025 are excluded from all data throughout this report. ## AI disclosure

- Typewriter effect with typo simulation was fully AI-orchestrated
- Co-occurrence hover interaction and sparkline tooltips were AI-assisted
- Data normalization pipeline was AI-assisted

Week 7 - March 30 - April 5, 2026

Unified world map

This was the most time-consuming change of the entire project, but absolutely worth it. I merged all separate map sections into a single unified world map where the user cycles through modes: co-occurrence, trend arrows, year heatmap, US city bubbles with metro sub-zooms into NYC, LA, and SF, and finally an exit credits page. The result feels like an app rather than a scrolling website, and I really like how it turned out.

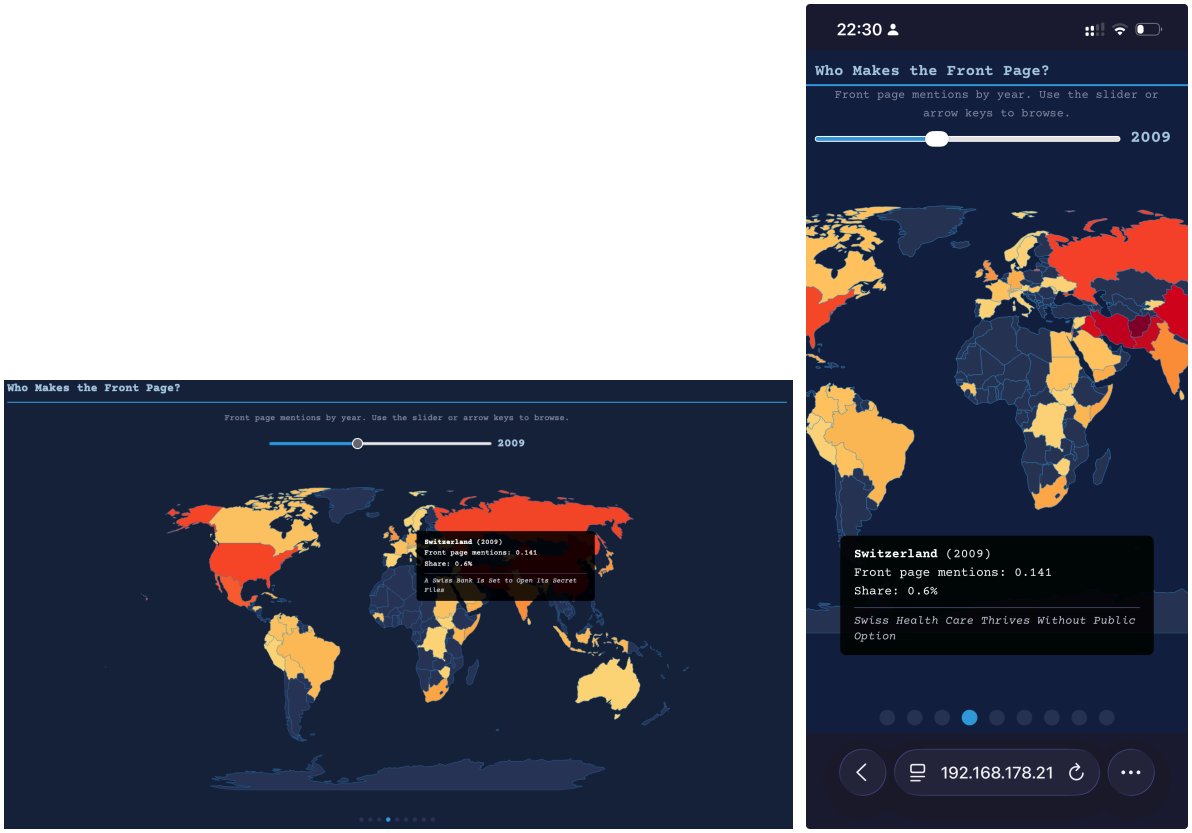
I also added a new year heatmap mode where each country is colored by its front page mention count for a selected year, using a choropleth with a slider to scrub through 2000-2023. The US city bubble map is now part of the world map too - when the user reaches it, the map smoothly zooms into the US by animating the SVG viewBox, and the city bubbles pop in with staggered transitions.

Title and exit overlays

The header and a credits page were integrated as the first and last modes. Both are fixed-position overlays on top of the map. The main difficulty was event propagation - the overlays block wheel events from reaching the map underneath, so I had to attach the same scroll handler to all overlay elements independently.

Mobile support

I added full touch support for mobile devices. On the title page, the user can swipe down or tap to enter. Navigation between modes uses tappable dots at the bottom of the screen. Countries and bubbles respond to tap instead of hover, with a tap-elsewhere-to-dismiss pattern. Tooltips appear as a fixed bar at the bottom of the screen, and the connection panel slides up as a bottom sheet.



Left: Desktop with hover tooltips. Right: Mobile with horizontal scroll and navigation dots.

The map is rendered wider than the viewport on mobile using a horizontal scroll container, so it has a usable size despite the portrait orientation. I also built a precision bubble selector for the US city map - the user toggles precision mode, then drags to aim at bubbles with the selection point offset above their finger, and the nearest bubble highlights. This was a lot more work than I initially thought it would be, but it works really well - the user can now precisely select even the smallest bubbles on a phone screen.

AI disclosure

- Title/exit overlay integration and event propagation fixes were AI-assisted
- Mobile touch support (swipe navigation, tap interactions, precision bubble selector) was AI-assisted

Week 8 - April 6-12, 2026

Trend range slider

I added a dual-handle range slider to the trend arrow map that lets the user select a custom year range (e.g. 2005-2015) instead of always computing trends over the full 2000-2023 period. The arrows recompute in real time as the range changes, which makes it possible to discover patterns that are invisible in the full-range view - like how Iraq's coverage spiked during the war years and declined after, or how Ukraine only becomes significant after 2014.

The slider also supports keyboard control: arrow keys move the end year, shift+arrow moves the start

year. The tooltip now shows a mini bar chart with the yearly front page share for the selected range and a dashed trendline.

Stable trend detection

Countries with very small trends relative to their own average coverage now get a gray horizontal arrow instead of a colored diagonal one, indicating "stable" coverage. The threshold is based on the ratio of the slope to the country's mean. I also discovered that the raw article counts were misleading because the NYT's total output dropped from ~110k articles/year in 2000 to ~50k in 2023. All trend data is now normalized as a percentage of total front page articles per year, which removes this distortion.

Missing country names

I discovered that several countries on the map were showing as numeric IDs (like "090" for Solomon Islands) because they were missing from the TopoJSON-to-ISO3 mapping. I added 32 missing country codes and a name fallback dictionary. I also found that the preprocessing script was missing several countries from its search dictionary (like Antarctica, Timor-Leste, and Solomon Islands among others). Both were fixed and the dataset was regenerated.

AI disclosure

- Trend range slider with dual handles and keyboard control was AI-assisted
- Stable trend detection logic was AI-assisted
- Missing country identification and preprocessing updates were AI-assisted

Peer Assessment

Solo project. All components - data preprocessing, visualization development, design decisions, mobile support, and process book - were implemented by a single person.

Sources

- NYT election arrow map inspiration: https://www.nytimes.com/interactive/2024/11/06/us/politics/presidential election-2024-red-shift.html
- Tile grid cartogram positions: https://github.com/mustafasaifee42/Tile-Grid-Map
- Dataset: NYT Articles: 2.1M+ (2000-Present), Kaggle